

# Instruction basics on the 2D fabric — the SAME opcode, Horizontal vs Vertical mode

One instruction per row. Colour tracks where data goes (numbers where the value matters). Flipping CSR 0x7c0 rotates each spatial op 90° — so a pixel is free to move anywhere on the grid. (vadd / vmv / vmerge are element-wise, so H = V.)

Inputs

Horizontal mode

Vertical mode (0x7c0=1)

## vmv.v.x (broadcast)

vmv.v.x v8, a0

a0 = 7

Copy one scalar into every lane.

7777

7777

7777

7777

every lane = 7

= same (H = V)

## vadd.vv

vadd.vv v8, vA, vB

Element-wise add. Position never moves.

1234

5678

2468

1357

+

4321

8765

8642

7531

5555

13131313

10101010

8888

A + B

= same (H = V)

## vslideup.vi vd, vs, 1

csrwi 0x7c0,h/v vslideup.vi v8, vs, 1

Shift by one. Mode picks the axis.

1234

5678

2468

1357

→

1234

5678

2468

1357

shift RIGHT (within row)

1234

5678

2468

1357

shift DOWN (between rows)

## vrgather

vrgather.vx v8, vs, idx

Read any lane → any lane. Pixels are not local!

1234

5678

2468

1357

→

1234

5678

2468

1357

H: top-left → top-right

1234

5678

2468

1357

V: top-right → bottom-right

## vle8 / vse8

csrwi 0x7c0,1 vle8.v v8,(a0)

Vertical load/store transposes the image.

1234

5678

2468

1357

→

1234

5678

2468

1357

H: copy through

1234

5678

2468

1357

V: transpose (rows ↔ cols)

## vredsum.vs

vmv.v.i v0,0 vredsum.vs v8, vs, v0

Add up. Mode picks the axis the sum lands on.

1234

1111

2020

4321

→

104410

· · · ·

· · · ·

· · · ·

H: row-sums in column 0

8686

· · · ·

· · · ·

· · · ·

V: col-sums in row 0

## vmerge.vvm + mask v0

vmerge.vvm v8, vB, vA, v0

Predicate per lane. Pick A where mask=1, else B.

A (teal)

B (pink)

mask v0

→

A where mask=1, else B

mask gates the SAME lanes in H & V; only the data orientation under it changes